

MiSo: Mixture-of-Submanifolds Nonnegative Matrix Factorization for Interpretable Factor-Usage Motifs

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Abstract

Nonnegative matrix factorization is widely used to represent a nonnegative data matrix through low-rank nonnegative factors and loadings. Many variants encourage individual observations to use only a small number of factors, but they usually do not model recurring patterns of factor usage as inferential objects. We introduce MiSo (Mixture of Submanifolds), a probabilistic model for count data that learns recurring factor-usage motifs. Each observation is assigned softly to a latent motif, and each motif defines a low-dimensional submanifold spanned by a small subset of shared factor rows. The factor rows themselves may be dense. Each motif dimension has a posterior distribution over the shared factor rows, giving both clustering by factor-usage motif and posterior uncertainty over the factors defining each motif. The method combines an initial Poisson NMF dictionary with row-wise Poisson SuSiE and a finite mixture model. We describe the model, a variational empirical-Bayes fitting procedure, and the simulation studies needed to evaluate factor recovery, motif recovery, overfitting behavior, and posterior uncertainty.

1 Introduction

Nonnegative matrix factorization (NMF) seeks a low-rank representation of a nonnegative matrix $Y \in \mathbb{N}^{N \times M}$ by writing

$$Y \approx LF,$$

where $L \in \mathbb{R}_+^{N \times K}$ contains nonnegative loadings and $F \in \mathbb{R}_+^{K \times M}$ contains nonnegative factors. In many scientific applications the rows of L are expected to be sparse: each observation should be explained by only a few latent factors. Sparse NMF methods address this by regularizing or constraining the loading matrix. However, sparsity itself is not the whole inferential target. Often we want to know whether many observations share the same kind of sparsity pattern, which factors define that pattern, and whether the evidence distinguishes one possible factor combination from another.

This paper develops MiSo, a mixture-of-submanifolds model for nonnegative count matrix factorization. The main idea is to treat recurring factor-usage patterns as latent motifs. Each motif represents a small collection of factor rows that tend to appear together in the loadings. Observations are softly assigned to motifs, and each motif has posterior uncertainty over which factors define its dimensions. Thus the model is designed to answer questions such as: Which factor-usage motifs are present? Which observations use each motif? Which factors define a motif? Are two nearly identical or correlated factors interchangeable in the posterior?

The statistical challenge is that sparse NMF has several sources of non-identifiability. Factor rows can be permuted, highly correlated factors can compete to explain the same counts, and an overfitted factorization can produce duplicate or redundant factors. A hard sparse loading estimate can hide this ambiguity by choosing one arbitrary support. We instead represent uncertainty with

posterior probabilities over factor identities, following the spirit of SuSiE [Wang et al., 2020]. The resulting posterior summaries are analogous to posterior inclusion probabilities, but they operate at the motif level rather than only at the observation level.

The paper has three methodological layers. Section 2 introduces Poisson SuSiE for one count vector and reviews finite mixture models. Section 3 introduces MiSo, which pools factor-usage patterns across observations through latent motifs. Section 4 is reserved for identifiability theory. Section 5 describes the experiments needed.

2 Background

MiSo combines two familiar constructions: a SuSiE representation with a Poisson observation model and a finite mixture model.

2.1 Poisson SuSiE model

A core problem in modeling a submanifold inside a large manifold is to choose the supported dimensions. We begin with a known nonnegative dictionary $F \in \mathbb{R}_+^{K \times M}$ and a count vector $y = (y_m)_{m=1}^M \in \mathbb{N}^M$ distributed as

$$y_m \stackrel{ind.}{\sim} \text{Poi}(\mu_m), \quad m = 1, \dots, M, \quad (1)$$

where $\mu = (\mu_m)_{m=1}^M$ is the unknown mean vector. Assume that μ is supported only on a D dimensional submanifold of a K dimensional manifold generated by rows $F_1, \dots, F_K \in \mathbb{R}^M$ of F . We model this by a SuSiE-style representation:

$$\mu = \sum_{d=1}^D \lambda_d F_{\gamma_d}, \quad (2)$$

where $\gamma_d \subset [K]$ choose factor rows and $\lambda_d \in \mathbb{R}_+$ is the loading size for dimension d . They are distributed a priori according to

$$\lambda_d \stackrel{ind.}{\sim} \text{Gam}(\alpha_d^0, \beta_d^0), \quad \gamma_d \stackrel{iid.}{\sim} \text{Cat}(1/K, \dots, 1/K), \quad d = 1, \dots, D. \quad (3)$$

where the Gamma distribution is parameterized by shape and rate. We make inference of the submanifold of y via the posterior distribution of $(\gamma_d)_{d=1}^D$. Following Gaussian SuSiE Wang et al. [2020], we approximate the posterior by mean-field family

$$q(\lambda, \gamma) = \prod_{d=1}^D q(\gamma_d) q(\lambda_d \mid \gamma_d), \quad (4)$$

which will be found by maximizing the following Evidence Lower Bound (ELBO). By the standard Jensen inequality, the model's evidence can be lower bounded as

$$\begin{aligned} \log p(y) &\geq \mathbb{E}_q[\log p(y)] - \text{KL}(q(\gamma, \lambda) \parallel p(\gamma, \lambda)) \\ &= \sum_{m=1}^M \mathbb{E}_q[\log p(y_m)] - \sum_{d=1}^D \text{KL}(q(\gamma_d, \lambda_d) \parallel p(\gamma_d, \lambda_d)) \\ &= \sum_{m=1}^M \mathbb{E}_q \left[\log y_m \log \left(\sum_{d=1}^D \lambda_d F_{\gamma_d m} \right) - \sum_{d=1}^D \lambda_d F_{\gamma_d m} \right] - \sum_{d=1}^D \text{KL}(q(\gamma_d, \lambda_d) \parallel p(\gamma_d, \lambda_d)) \end{aligned}$$

where second line uses the independency of elements of y and of the mean-field family q , and the third line is up to a constant ($y_m!$) only depends on data.

$$q(\gamma_d = k) = \bar{\gamma}_{dk}, \quad q(\lambda_d \mid \gamma_d = k) = \text{Gam}(\alpha_{dk}, \beta_{dk}). \quad (5)$$

We also introduce allocation variables ξ_{dm} satisfying $\xi_{dm} \geq 0$ and $\sum_{d=1}^D \xi_{dm} = 1$ for each coordinate m . They allocate the observed count y_m across the D dimensions. Jensen's inequality gives the lower bound contribution

$$y_m \log \left(\sum_{d=1}^D \lambda_d F_{\gamma_{dm}} \right) \geq \sum_{d=1}^D \xi_{dm} y_m \log \left(\frac{\lambda_d F_{\gamma_{dm}}}{\xi_{dm}} \right).$$

The resulting coordinate updates have a simple form. The allocation update is

$$\xi_{dm} \propto \exp \left[\sum_{k=1}^K \bar{\gamma}_{dk} \{ \psi(\alpha_{dk}) - \log \beta_{dk} \} + \sum_{k=1}^K \bar{\gamma}_{dk} \log F_{km} \right], \quad (6)$$

normalized over $d = 1, \dots, D$. The Gamma variational parameters are

$$\alpha_{dk} \leftarrow \alpha_d^0 + \sum_{m=1}^M \xi_{dm} y_m, \quad \beta_{dk} \leftarrow \beta_d^0 + \sum_{m=1}^M F_{km}. \quad (7)$$

The posterior probability that dimension d uses factor k is updated by a softmax over the terms in the lower bound depending on $\gamma_d = k$:

$$\bar{\gamma}_{dk} \propto \exp \left[\sum_{m=1}^M \xi_{dm} y_m \log F_{km} - \left(\alpha_d^0 + \sum_{m=1}^M \xi_{dm} y_m \right) \log \left(\beta_d^0 + \sum_{m=1}^M F_{km} \right) \right]. \quad (8)$$

When the rows of F are normalized, the rate term is constant across k , but we keep it visible because it clarifies the role of the Gamma prior and the scale convention.

Empirical-Bayes prior update. Poisson SuSiE can use empirical Bayes to update the prior mean of each loading dimension. For dimension d , define the posterior mean and posterior mean log-size

$$\sum_{k=1}^K \bar{\gamma}_{dk} \frac{\alpha_{dk}}{\beta_{dk}}, \quad \sum_{k=1}^K \bar{\gamma}_{dk} \{ \psi(\alpha_{dk}) - \log \beta_{dk} \}.$$

Maximizing the expected Gamma prior gives the rate update

$$\beta_d^0 = \frac{\alpha_d^0}{\sum_{k=1}^K \bar{\gamma}_{dk} \alpha_{dk} / \beta_{dk}},$$

and α_d^0 solves

$$\psi(\alpha_d^0) = \sum_{k=1}^K \bar{\gamma}_{dk} \{ \psi(\alpha_{dk}) - \log \beta_{dk} \} + \log \beta_d^0.$$

This empirical-Bayes step is important when D is overspecified: dimensions with little evidence can have prior mean α_d^0 / β_d^0 close to zero, which is an automatic relevance determination (ARD) mechanism for selecting the effective number of dimensions.

Algorithm 1 Poisson SuSiE with a fixed dictionary

Require: Count vector y , row-normalized dictionary F , nominal dimension D , initial Gamma parameters (α_d^0, β_d^0) , maximum iterations T , and tolerance δ .

- 1: Initialize every row of $\bar{\gamma}$ to a positive probability vector, using a small random perturbation to break symmetry.
 - 2: Set $\xi_{dm} = 1/D$, $\alpha_{dk} = \alpha_d^0 + D^{-1} \sum_m y_m$, and $\beta_{dk} = \beta_d^0 + \sum_m F_{km}$.
 - 3: **for** $t = 1, \dots, T$ **do**
 - 4: **for** $d = 1, \dots, D$ **do**
 - 5: Update all $\xi_{d'm}$ using Eq. (6) and normalize over $d' = 1, \dots, D$.
 - 6: Update α_{dk} and β_{dk} for every k using Eq. (7).
 - 7: Update $\bar{\gamma}_{dk}$ for every k using Eq. (8) and normalize over k .
 - 8: **if** empirical Bayes is enabled **then**
 - 9: Update (α_d^0, β_d^0) as described below.
 - 10: **end if**
 - 11: **end for**
 - 12: Evaluate the augmented variational lower bound.
 - 13: **if** its relative improvement is less than δ **then**
 - 14: **break**
 - 15: **end if**
 - 16: **end for**
 - 17: **return** $\bar{\gamma}$, α , β , ξ , and $\tilde{L}_k = \sum_d \bar{\gamma}_{dk} \alpha_{dk} / \beta_{dk}$.
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2.2 Finite mixture models

Let $Z_i \in \{1, \dots, S\}$ denote the component of observation Y_i , with $\Pr(Z_i = s) = \pi_s$. If component s has likelihood $p_s(Y_i)$, then

$$p(Y_i) = \sum_{s=1}^S \pi_s p_s(Y_i).$$

Suppose $\mathcal{L}_{is}$ is a lower bound on $\log p_s(Y_i)$. A variational mixture update assigns responsibility

$$\omega_{is} = q(Z_i = s) = \frac{\pi_s \exp(\mathcal{L}_{is})}{\sum_{r=1}^S \pi_r \exp(\mathcal{L}_{ir})}, \quad \pi_s \leftarrow \frac{1}{N} \sum_{i=1}^N \omega_{is}.$$

MiSo uses a Poisson SuSiE component for each motif and shares each motif's factor-selection variables across the observations assigned to that component.

3 MiSo

3.1 Generative model

MiSo models repeated factor-usage motifs. The observed data are counts $Y \in \mathbb{N}^{N \times M}$. The potentially dense factor matrix $F \in \mathbb{R}_+^{K \times M}$ is row-normalized as above. Each motif identifies a low-dimensional submanifold spanned by a subset of these shared factor rows; the model does not impose sparsity on F . There are S latent motifs. We initially fit each motif with a nominal dimension D , but the effective dimension is allowed to be smaller and can vary by motif.

For motif s and dimension d , the factor identity is random:

$$\gamma_{sd} \sim \text{Cat}(\rho_{sd1}, \dots, \rho_{sdK}).$$

For observation i ,

$$Z_i \sim \text{Cat}(\pi_1, \dots, \pi_S).$$

Conditional on $Z_i = s$, the observation has one nonnegative loading size for each dimension:

$$\lambda_{id} \mid Z_i = s \sim \text{Gam}(\alpha_{sd}^0, \beta_{sd}^0), \quad d = 1, \dots, D.$$

The observed counts are

$$Y_{im} \mid Z_i = s, \lambda_i, \gamma, F \sim \text{Poi}(\mu_{im}), \quad \mu_{im} = \sum_{d=1}^D \lambda_{id} F_{\gamma_{sd}m}, \quad m = 1, \dots, M.$$

Equivalently, if L_i denotes the loading vector for observation i , then

$$L_i = \sum_{d=1}^D \lambda_{id} e_{\gamma_{Z_i d}},$$

where e_k is the k th coordinate vector in $\mathbb{R}^K$. Thus all observations assigned to motif s share the same motif-level factor identities $\gamma_{s1}, \dots, \gamma_{sD}$, but each observation has its own loading sizes.

3.2 Variational family

The variational approximation is

$$q(Z, \gamma, \lambda) = \prod_{i=1}^N q(Z_i) \prod_{s=1}^S \prod_{d=1}^D q(\gamma_{sd}) \prod_{i=1}^N \prod_{s=1}^S \prod_{d=1}^D q(\lambda_{id} \mid Z_i = s).$$

We write

$$q(Z_i = s) = \omega_{is}, \quad q(\gamma_{sd} = k) = \bar{\gamma}_{sdk},$$

and

$$q(\lambda_{id} \mid Z_i = s) = \text{Gam}(\alpha_{isd}, \beta_{sd}).$$

The rate parameter β_{sd} does not depend on i because the Poisson factor rows are shared and row-normalized; the shape parameter α_{isd} depends on the counts allocated from observation i .

For each candidate motif s , we introduce allocation variables $\xi_{isd m}$ satisfying

$$\xi_{isd m} \geq 0, \quad \sum_{d=1}^D \xi_{isd m} = 1.$$

They allocate Y_{im} across the dimensions of motif s . The implementation computes $\xi_{isd m}$ in feature blocks rather than storing the full $N \times S \times D \times M$ tensor.

3.3 Coordinate updates

The allocation update is

$$\xi_{isd m} \propto \exp \left[\psi(\alpha_{isd}) - \log \beta_{sd} + \sum_{k=1}^K \bar{\gamma}_{sdk} \log F_{km} \right], \quad (9)$$

normalized over $d = 1, \dots, D$. The Gamma posterior update is

$$\alpha_{isd} \leftarrow \alpha_{sd}^0 + \sum_{m=1}^M \xi_{isd m} Y_{im}, \quad \beta_{sd} \leftarrow \beta_{sd}^0 + \sum_{m=1}^M \sum_{k=1}^K \bar{\gamma}_{sd k} F_{km}. \quad (10)$$

The second term is equal to one when each row of F is normalized, but we write the full expression to make the model dependence explicit.

For motif dimension (s, d) , the posterior probability of factor k is updated by

$$\bar{\gamma}_{sd k} \propto \rho_{sd k} \exp \left[\sum_{i=1}^N \omega_{is} \left\{ \sum_{m=1}^M \xi_{isd m} Y_{im} \log F_{km} - \frac{\alpha_{isd}}{\beta_{sd}} \sum_{m=1}^M F_{km} \right\} \right]. \quad (11)$$

This is the central posterior uncertainty object in MiSo. A concentrated vector $\bar{\gamma}_{sd1}, \dots, \bar{\gamma}_{sdK}$ means that motif dimension (s, d) has a well-identified factor. A diffuse vector means that several factor rows are plausible explanations for the same motif dimension.

The motif responsibility for observation i is updated from the component lower bound for that observation under each motif:

$$\omega_{is} \propto \pi_s \exp\{\mathcal{L}_{is}\}, \quad (12)$$

where

$$\begin{aligned} \mathcal{L}_{is} = & \sum_{d=1}^D \sum_{m=1}^M \xi_{isd m} Y_{im} \left\{ \psi(\alpha_{isd}) - \log \beta_{sd} + \sum_{k=1}^K \bar{\gamma}_{sd k} \log F_{km} - \log \xi_{isd m} \right\} \\ & - \sum_{d=1}^D \sum_{m=1}^M \frac{\alpha_{isd}}{\beta_{sd}} \sum_{k=1}^K \bar{\gamma}_{sd k} F_{km} - \sum_{d=1}^D \text{KL}\{\text{Gam}(\alpha_{isd}, \beta_{sd}) \parallel \text{Gam}(\alpha_{sd}^0, \beta_{sd}^0)\}. \end{aligned}$$

The mixture weights are updated by

$$\pi_s \leftarrow \frac{1}{N} \sum_{i=1}^N \omega_{is}. \quad (13)$$

3.4 Dictionary estimation and optional refinement

When the dictionary is unknown, we first fit Poisson NMF to obtain $Y \approx L^{\text{NMF}} F^{\text{NMF}}$. We row-normalize F^{NMF} , absorbing each removed row scale into the corresponding column of L^{NMF} , and denote the resulting dictionary by $\hat{F}$. This dictionary is frozen while Poisson SuSiE is fitted independently to the rows of Y for initialization. Thus the initialization stage never alternates between the SuSiE posteriors and the factor dictionary.

The main MiSo model can either retain $F = \hat{F}$ or refine it after MiSo has been initialized. If refinement is enabled, the expected allocated counts are

$$A_{km} = \sum_{i=1}^N \sum_{s=1}^S \sum_{d=1}^D \omega_{is} \bar{\gamma}_{sd k} \xi_{isd m} Y_{im}.$$

The row-normalized update is

$$F_{km} \leftarrow \frac{A_{km} + \epsilon}{\sum_{m'=1}^M (A_{km'} + \epsilon)}. \quad (14)$$

Keeping F fixed isolates motif learning from dictionary estimation; updating F allows the dictionary to adapt to the pooled motif structure.

3.5 Initialization

Initialization is an important part given the non-concave nature of the optimization problem. There are many ways to initialize parameters in MiSo, depending on VI blocks.

My attempt. With the Poisson NMF dictionary $\hat{F}$ frozen, we apply Algorithm 1 independently to each row Y_i . Let $\bar{\gamma}_{idk}^{\text{PS}}$, α_{idk}^{PS} , and β_{idk}^{PS} denote the resulting row-wise Poisson SuSiE parameters. Our goal is to initialize ω_{is} , π_s , $\bar{\gamma}_{sdk}$, α_{sd}^0 , and β_{sd}^0 from these. After these quantities have been initialized, construct the local quantities internally:

$$\alpha_{isd} = \alpha_{sd}^0 + \frac{\sum_m Y_{im}}{D},$$

$$\beta_{sd} = \beta_{sd}^0 + \sum_k \bar{\gamma}_{sdk} \sum_m \hat{F}_{km},$$

followed by

$$\xi_{isd} \propto \exp \left\{ \mathbb{E} \log \lambda_{isd} + \sum_k \bar{\gamma}_{sdk} \log \hat{F}_{km} \right\}.$$

We aim to create $\mathcal{A}_i \subset [K]$ that captures the major factors supporting observation i , with $|\mathcal{A}_i| \leq D$:

1. Set

$$\text{PIP}_{ik} = 1 - \prod_{d=1}^D (1 - \bar{\gamma}_{idk}^{\text{PS}}), \quad k \in [K]. \quad (15)$$

2. Let $b_{i1}, \dots, b_{iK}$ order these PIPs from largest to smallest. For $\tau = \text{init_pip_truncate_level}$, define

$$r_i = \min \left\{ r : \frac{\sum_{j=1}^r \text{PIP}_{i,b_{ij}}}{\sum_{k=1}^K \text{PIP}_{ik}} \geq \tau \right\}.$$

We use $\tau = 0.95$ by default.

3. Set

$$\mathcal{A}_i = \{b_{i1}, \dots, b_{i,\min(r_i, D)}\}.$$

Thus $\mathcal{A}_i$ contains at most D factors, but it may contain fewer.

Let S^* be the number of unique sets among $\{\mathcal{A}_i : i \in [N]\}$, let $\mathcal{A}_1^*, \dots, \mathcal{A}_{S^*}^*$ be those distinct sets, and define

$$\mathcal{N}_s^* = \{i : \mathcal{A}_i = \mathcal{A}_s^*\}.$$

1. **Let's test this first.** If S is not specified, set $S = S^*$. Thus, for fixed D , the user only needs to specify `init_pip_truncate_level`. For $i \in [N]$ and $s \in [S]$, initialize the responsibilities and mixture weights by

$$\omega_{is}^{(0)} = \mathbf{1}\{i \in \mathcal{N}_s^*\}, \quad \pi_s^{(0)} = \frac{|\mathcal{N}_s^*|}{N}. \quad (16)$$

Within cluster s , average the row-wise PIPs,

$$\overline{\text{PIP}}_{sk} = \frac{1}{|\mathcal{N}_s^*|} \sum_{i \in \mathcal{N}_s^*} \text{PIP}_{ik}, \quad k \in [K]. \quad (17)$$

Let $a_{s1}, \dots, a_{sK}$ be a permutation of $1, \dots, K$ such that

$$\overline{\text{PIP}}_{s,a_{s1}} \geq \dots \geq \overline{\text{PIP}}_{s,a_{sK}}.$$

Assuming $D \leq K$, use the first D factors as distinct anchors and set

$$\bar{\gamma}_{sdk}^{(0)} = (1 - \varepsilon_\gamma) \mathbf{1}\{k = a_{sd}\} + \frac{\varepsilon_\gamma}{K}, \quad d \in [D], \quad k \in [K]. \quad (18)$$

We use $\varepsilon_\gamma = 0.05$ by default. This expands every cluster to the common nominal dimension D , while the ordering of all D slots is determined by the average PIP within that cluster.

The Poisson SuSiE slot labels are exchangeable across observations. Therefore, before initializing the Gamma priors, align the row-wise slots to the ordered cluster anchors. For each $i \in \mathcal{N}_s^*$, choose

$$p_i = \arg \max_{p \in \mathfrak{S}_D} \sum_{d=1}^D \bar{\gamma}_{i,p(d),a_{sd}}^{\text{PS}} \frac{\alpha_{i,p(d),a_{sd}}^{\text{PS}}}{\beta_{i,p(d),a_{sd}}^{\text{PS}}}, \quad (19)$$

where $\mathfrak{S}_D$ is the set of permutations of $1, \dots, D$. This is a linear assignment problem, so it can be solved without enumerating the $D!$ permutations.

Directly averaging Gamma shapes and rates is generally not appropriate: different shape-rate pairs are on a nonlinear scale and their arithmetic averages need not represent the average posterior distribution. Instead, average the two sufficient posterior moments after slot alignment:

$$\bar{\lambda}_{sd}^{\text{PS}} = \frac{1}{|\mathcal{N}_s^*|} \sum_{i \in \mathcal{N}_s^*} \sum_{k=1}^K \bar{\gamma}_{i,p_i(d),k}^{\text{PS}} \frac{\alpha_{i,p_i(d),k}^{\text{PS}}}{\beta_{i,p_i(d),k}^{\text{PS}}}, \quad (20)$$

$$\overline{\log \lambda}_{sd}^{\text{PS}} = \frac{1}{|\mathcal{N}_s^*|} \sum_{i \in \mathcal{N}_s^*} \sum_{k=1}^K \bar{\gamma}_{i,p_i(d),k}^{\text{PS}} \left\{ \psi\left(\alpha_{i,p_i(d),k}^{\text{PS}}\right) - \log \beta_{i,p_i(d),k}^{\text{PS}} \right\}. \quad (21)$$

Because $\hat{F}$ is row-normalized, the Poisson SuSiE updates make α_{idk}^{PS} and β_{idk}^{PS} constant in k for fixed (i, d) . In that case, the sums over k above simplify to the posterior mean and posterior mean log-size of the aligned slot, $\alpha_{i,p_i(d)}^{\text{PS}}/\beta_{i,p_i(d)}^{\text{PS}}$ and $\psi(\alpha_{i,p_i(d)}^{\text{PS}}) - \log \beta_{i,p_i(d)}^{\text{PS}}$, respectively. Initialize α_{sd}^0 and β_{sd}^0 by matching these moments to a Gamma distribution:

$$\beta_{sd}^0 = \frac{\alpha_{sd}^0}{\bar{\lambda}_{sd}^{\text{PS}}}, \quad (22)$$

$$\psi(\alpha_{sd}^0) - \log \beta_{sd}^0 = \overline{\log \lambda}_{sd}^{\text{PS}}. \quad (23)$$

Equivalently, solve the one-dimensional equation

$$\log \alpha_{sd}^0 - \psi(\alpha_{sd}^0) = \log \bar{\lambda}_{sd}^{\text{PS}} - \overline{\log \lambda}_{sd}^{\text{PS}},$$

and then use Eq. (22). A small positive numerical floor should be applied to $\bar{\lambda}_{sd}^{\text{PS}}$ in the implementation. These equations use every row-wise factor possibility rather than only the anchor, while the alignment in Eq. (19) ensures that the same motif dimension is averaged across observations. This is precisely one MiSo empirical-Bayes prior update with hard initial cluster responsibilities and the row-wise Poisson SuSiE posteriors substituted for the MiSo local posteriors.

2. If S is specified and $S^* \geq S$, let $\mathcal{N}_s^* = \{i : \mathcal{A}_i = \mathcal{A}_s^*\}$. Reorder the distinct sets so that $(|\mathcal{N}_s^*|)_{s=1}^{S^*}$ is decreasing, and retain the first S sets. For $s \in [S]$, initialize the factor-selection probabilities by

$$\bar{\gamma}_{sdk}^{(0)} = \frac{1}{|\mathcal{N}_s^*|} \sum_{i \in \mathcal{N}_s^*} \bar{\gamma}_{idk}^{\text{PS}}, \quad d \in [D], \quad k \in [K]. \quad (24)$$

For $i \in [N]$ and $s \in [S]$, initialize the responsibilities by

$$\omega_{is}^{(0)} = \begin{cases} \mathbf{1}\{\mathcal{A}_i = \mathcal{A}_s^*\}, & \mathcal{A}_i \in \{\mathcal{A}_1^*, \dots, \mathcal{A}_S^*\}, \\ 1/S, & \mathcal{A}_i \in \{\mathcal{A}_{S+1}^*, \dots, \mathcal{A}_{S^*}^*\}. \end{cases} \quad (25)$$

Thus observations belonging to one of the retained sets receive a hard initial assignment, whereas observations belonging to a discarded set are initially distributed uniformly across the retained submanifolds.

3. If S is specified and $S > S^*$: Just set $S = S^*$ and proceed. We will go back to this if the current solution is not good enough.

Codex's suggestion. With the Poisson NMF dictionary $\hat{F}$ frozen, we apply Algorithm 1 independently to each row Y_i . Let $\bar{\gamma}_{idk}^{\text{PS}}$, α_{idk}^{PS} , and β_{idk}^{PS} denote the resulting row-wise Poisson SuSiE parameters. The preliminary expected loading of factor k in observation i is

$$\tilde{L}_{ik} = \sum_{d=1}^D \bar{\gamma}_{idk}^{\text{PS}} \frac{\alpha_{idk}^{\text{PS}}}{\beta_{idk}^{\text{PS}}}.$$

We remove observation-level scale before clustering,

$$x_{ik} = \frac{\tilde{L}_{ik}}{\sum_{h=1}^K \tilde{L}_{ih}}, \quad x_i \in \Delta^{K-1},$$

and obtain S preliminary centers by solving

$$(\hat{z}, \hat{c}) = \arg \min_{z_1, \dots, z_N, c_1, \dots, c_S} \sum_{i=1}^N \|x_i - c_{z_i}\|_2^2, \quad c_s \in \Delta^{K-1}.$$

For preliminary cluster s , let

$$r_s(1), \dots, r_s(K)$$

order the factors so that

$$c_{s, r_s(1)} \geq c_{s, r_s(2)} \geq \dots \geq c_{s, r_s(K)}.$$

Assuming $D \leq K$, MiSo uses the distinct top- D anchors

$$\kappa_{sd}^{(0)} = r_s(d), \quad d = 1, \dots, D, \quad (26)$$

to define D alignment anchors. The Poisson SuSiE slot labels are exchangeable across observations, so they must be aligned before their posteriors can be averaged. For each observation with $\hat{z}_i = s$, choose the permutation

$$p_i = \arg \max_{p \in \mathfrak{S}_D} \sum_{d=1}^D \bar{\gamma}_{i, p(d), \kappa_{sd}^{(0)}}^{\text{PS}} \frac{\alpha_{i, p(d), \kappa_{sd}^{(0)}}^{\text{PS}}}{\beta_{i, p(d), \kappa_{sd}^{(0)}}^{\text{PS}}}, \quad (27)$$

where $\mathfrak{S}_D$ is the set of permutations of $1, \dots, D$. This is a linear assignment problem. It matches each motif slot to one row-wise Poisson SuSiE slot using the posterior expected loading assigned to the corresponding anchor factor.

To prevent observations with large total counts from dominating the initialization, define the within-observation loading share of Poisson SuSiE slot d by

$$w_{id} = \frac{\sum_{k=1}^K \bar{\gamma}_{idk}^{\text{PS}} \alpha_{idk}^{\text{PS}} / \beta_{idk}^{\text{PS}}}{\sum_{d'=1}^D \sum_{k=1}^K \bar{\gamma}_{id'k}^{\text{PS}} \alpha_{id'k}^{\text{PS}} / \beta_{id'k}^{\text{PS}}}. \quad (28)$$

The cluster-averaged posterior for motif slot (s, d) is then

$$q_{sdk}^{(0)} = \frac{\sum_{i: \hat{z}_i = s} w_{i, p_i(d)} \bar{\gamma}_{i, p_i(d), k}^{\text{PS}}}{\sum_{i: \hat{z}_i = s} w_{i, p_i(d)}}. \quad (29)$$

If the denominator is numerically zero, we instead set $q_{sdk}^{(0)} = \mathbf{1}\{k = \kappa_{sd}^{(0)}\}$. Finally, for a small $\varepsilon_\gamma > 0$, initialize

$$\bar{\gamma}_{sdk}^{(0)} = (1 - \varepsilon_\gamma) q_{sdk}^{(0)} + \frac{\varepsilon_\gamma}{K}, \quad k = 1, \dots, K. \quad (30)$$

We use $\varepsilon_\gamma = 0.05$ by default. Unlike an initialization that replaces every slot by a point mass on its anchor, Eq. (30) retains the factor uncertainty learned by row-wise Poisson SuSiE. The distinct anchors only resolve slot-label switching; they do not replace the averaged posterior.

The initial mixture weights and responsibilities are set to

$$\pi_s^{(0)} = \frac{1}{S}, \quad \omega_{is}^{(0)} = \frac{1}{S}.$$

The preliminary partition is therefore used to initialize motif factor posteriors but is not imposed as a hard assignment. The Gamma variational and prior parameters are initialized symmetrically across motifs and are then updated using the raw count matrix.

3.6 Empirical-Bayes dimension selection

For motif s and dimension d , define the responsibility-weighted posterior moments

$$\begin{aligned} \bar{\lambda}_{sd} &= \frac{\sum_{i=1}^N \omega_{is} \alpha_{isd} / \beta_{sd}}{\sum_{i=1}^N \omega_{is}}, \\ \overline{\log \lambda}_{sd} &= \frac{\sum_{i=1}^N \omega_{is} \{\psi(\alpha_{isd}) - \log \beta_{sd}\}}{\sum_{i=1}^N \omega_{is}}. \end{aligned}$$

The empirical-Bayes update is

$$\beta_{sd}^0 = \frac{\alpha_{sd}^0}{\bar{\lambda}_{sd}}, \quad \psi(\alpha_{sd}^0) = \overline{\log \lambda}_{sd} + \log \beta_{sd}^0, \quad (31)$$

where the second equation is solved numerically. Each fitted motif starts with the same nominal dimension D , but its effective dimension can be smaller. The empirical-Bayes Gamma prior mean

$$\frac{\alpha_{sd}^0}{\beta_{sd}^0}$$

Algorithm 2 MiSo variational empirical-Bayes fitting

Require: Count matrix Y , factor count K , nominal motif dimension D , PIP truncation level τ , inner iterations R , outer iterations T , tolerance δ , and a choice of whether to refine F .

- 1: Fit rank- K Poisson NMF; row-normalize its factor matrix to obtain $\hat{F}$ and hold $\hat{F}$ fixed during the next two steps.
 - 2: **for** $i = 1, \dots, N$ **do**
 - 3: Apply Algorithm 1 to Y_i with dictionary $\hat{F}$; retain $\bar{\gamma}_{idk}^{\text{PS}}$, α_{idk}^{PS} , and β_{idk}^{PS} .
 - 4: **end for**
 - 5: Compute Eq. (15); truncate each ordered PIP profile at level τ to obtain $\mathcal{A}_i$, and set S to the number of distinct sets among $\mathcal{A}_1, \dots, \mathcal{A}_N$.
 - 6: Initialize ω_{is} and π_s using Eq. (16); order the cluster-average PIPs using Eq. (17).
 - 7: Initialize $\bar{\gamma}_{sdk}$ from the distinct top- D anchors using Eq. (18), and align the row-wise Poisson SuSiE slots using Eq. (19).
 - 8: Initialize $(\alpha_{sd}^0, \beta_{sd}^0)$ by the posterior-moment averages in Eqs. (20)–(23).
 - 9: Set $F = \hat{F}$, set $\xi_{isdm} = 1/D$, and initialize $\alpha_{isd} = \alpha_{sd}^0 + D^{-1} \sum_m Y_{im}$ and β_{sd} using Eq. (10).
 - 10: **for** $t = 1, \dots, T$ **do**
 - 11: **for** $r = 1, \dots, R$ **do**
 - 12: Update ξ_{isdm} for all i, s, d, m using Eq. (9), processing features in blocks.
 - 13: Update α_{isd} and β_{sd} using Eq. (10).
 - 14: **end for**
 - 15: Evaluate $\mathcal{L}_{is}$ for every observation and candidate motif.
 - 16: Update ω_{is} and π_s using Eqs. (12) and (13).
 - 17: **if** empirical Bayes is enabled **then**
 - 18: Update $(\alpha_{sd}^0, \beta_{sd}^0)$ using Eq. (31).
 - 19: **end if**
 - 20: Update $\bar{\gamma}_{sdk}$ using Eq. (11); optionally damp the update.
 - 21: **if** dictionary refinement is enabled **then**
 - 22: Update F using Eq. (14); optionally damp the update and row-normalize.
 - 23: **end if**
 - 24: Evaluate the mixture variational objective.
 - 25: **if** its relative improvement is less than δ **then**
 - 26: **break**
 - 27: **end if**
 - 28: **end for**
 - 29: Recompute the local posteriors and responsibilities using the final global parameters.
 - 30: **return** ω , π , $\bar{\gamma}$, α , β_{sd} , $\hat{D}_s$, and F .
-

estimates the loading size expected for that dimension. If this value is close to zero relative to the other dimensions in motif s , then dimension d is effectively inactive. This gives an ARD-style selected dimension

$$\hat{D}_s = \# \left\{ d : \frac{\alpha_{sd}^0 / \beta_{sd}^0}{\sum_{d'=1}^D \alpha_{sd'}^0 / \beta_{sd'}^0} \geq \tau \right\},$$

for a threshold τ .

This ARD diagnostic should be interpreted separately from the entropy of $\bar{\gamma}_{sd1}, \dots, \bar{\gamma}_{sdK}$. The ARD prior mean asks whether a dimension carries loading mass. The entropy of $\bar{\gamma}_{sd}$ asks whether the factor identity of that dimension is uncertain. Both are useful: an inactive dimension should have small loading mass, while an active but ambiguous dimension may have large loading mass and diffuse posterior factor identity.

4 Identifiability Theory

MiSo places its geometric structure on the latent Poisson intensity, not on the observed count vector. This distinction lets us retain the language of submanifolds while using cones when a closed support or an extreme ray is the mathematically relevant object. For a realized motif s , write

$$H_s = \sum_{d=1}^{D_s} a_{sd} \delta_{\theta_{sd}}, \quad \theta_{sd} = \frac{F_{\gamma_{sd}}}{\beta_{sd}}, \quad a_{sd}, \beta_{sd} > 0, \quad (32)$$

where a_{sd} and β_{sd} are the Gamma shape and rate. Coincident atoms are combined by adding their shapes, giving the canonical representation

$$H = \sum_{j=1}^J a_j \delta_{\theta_j}, \quad a_j > 0, \quad \theta_1, \dots, \theta_J \text{ pairwise distinct.} \quad (33)$$

The corresponding latent submanifold and its closure cone are

$$\mathcal{M}(H) = \left\{ \sum_{j=1}^J x_j \theta_j : x_j > 0 \right\}, \quad (34)$$

$$\mathcal{C}(H) = \overline{\mathcal{M}(H)} = \text{cone}\{\theta_1, \dots, \theta_J\}. \quad (35)$$

Thus $\mathcal{M}(H)$ is the relative interior of $\mathcal{C}(H)$ and is a smooth submanifold of its linear span. Its intrinsic dimension is $\dim \mathcal{M}(H) = \text{rank}(\theta_1, \dots, \theta_J)$, which can be smaller than the nominal number of dimensions D_s .

To define the distribution encoded by H , independently draw

$$\xi_j \sim \text{Gam}(a_j, 1), \quad j = 1, \dots, J,$$

and set

$$\mu_H = \sum_{j=1}^J \xi_j \theta_j. \quad (36)$$

We denote the law of μ_H by $\text{Gam}(H)$. This agrees with the original shape–rate parameterization because, if $\lambda_j \sim \text{Gam}(a_j, \beta_j)$, then $\beta_j \lambda_j \sim \text{Gam}(a_j, 1)$ and $\lambda_j F_{\gamma_j} = (\beta_j \lambda_j)(F_{\gamma_j}/\beta_j)$. Conditional on μ_H , the coordinates of Y are independent with

$$Y_m \mid \mu_H \sim \text{Poi}((\mu_H)_m), \quad m = 1, \dots, M.$$

Although $\mu_H \in \mathcal{M}(H)$ almost surely, the topological support of its distribution is the closure cone $\mathcal{C}(H)$.

The population mixture of latent submanifolds is represented by

$$G = \sum_{s=1}^S \pi_s \delta_{H_s}, \quad \pi_s > 0, \quad \sum_{s=1}^S \pi_s = 1, \quad (37)$$

and its observed count law is

$$\text{MiSo}(G) = \sum_{s=1}^S \pi_s \text{Poi}\{\text{Gam}(H_s)\}. \quad (38)$$

The motif measure H_s need not be a probability measure: its atom masses are Gamma shapes, whereas π_s is the mixture probability. For $t \in \mathbb{R}_+^M$, define

$$\Phi_H(t) := \mathbb{E} \exp\{-\langle t, \mu_H \rangle\} = \prod_{j=1}^J (1 + \langle \theta_j, t \rangle)^{-a_j}. \quad (39)$$

The probability-generating function of the noisy counts is

$$\mathbb{E}_H \left[\prod_{m=1}^M z_m^{Y_m} \right] = \Phi_H(\mathbf{1} - z), \quad z \in [0, 1]^M. \quad (40)$$

This identity is the bridge from the observed Poisson law to the geometry of the latent submanifolds.

Theorem 1 (Identification of one latent submanifold). *Let H and H' be finite positive atomic measures on $\mathbb{R}_+^M \setminus \{0\}$. Then*

$$\text{Poi}\{\text{Gam}(H)\} \stackrel{d}{=} \text{Poi}\{\text{Gam}(H')\} \implies H = H'.$$

Consequently, the latent submanifold $\mathcal{M}(H)$, its closure cone $\mathcal{C}(H)$, and its intrinsic dimension are identifiable from the population distribution of the Poisson observations.

Proof. Equality of the count laws gives equality of their probability-generating functions. By (40), $\Phi_H(t) = \Phi_{H'}(t)$ for $t \in (0, 1)^M$. Hence their negative log-Laplace transforms agree:

$$\sum_j a_j \log(1 + \langle \theta_j, t \rangle) = \sum_\ell a'_\ell \log(1 + \langle \theta'_\ell, t \rangle).$$

Differentiating on this open set gives

$$\sum_j \frac{a_j \theta_j}{1 + \langle \theta_j, t \rangle} = \sum_\ell \frac{a'_\ell \theta'_\ell}{1 + \langle \theta'_\ell, t \rangle}. \quad (41)$$

Choose $v \in \mathbb{R}_{++}^M$ such that distinct vectors among the finitely many θ_j and θ'_ℓ have distinct projections onto v . Such a v exists because the excluded choices lie in finitely many proper hyperplanes. Put $r_j = \langle \theta_j, v \rangle$ and $r'_\ell = \langle \theta'_\ell, v \rangle$, all of which are positive. On the real line $t = xv$, equation (41) becomes

$$\sum_j \frac{a_j \theta_j / r_j}{x + 1/r_j} = \sum_\ell \frac{a'_\ell \theta'_\ell / r'_\ell}{x + 1/r'_\ell}$$

for x in a positive interval. Multiplication by all denominators gives an identity between vector-valued real polynomials. Therefore the two real partial-fraction decompositions have the same denominator roots and the same coefficient at each root. For a matched pair with $r_j = r'_\ell$, multiplying by $x + 1/r_j$ and taking the real limit $x \rightarrow -1/r_j$ gives

$$\frac{a_j \theta_j}{r_j} = \frac{a'_\ell \theta'_\ell}{r_j}.$$

Taking the inner product with v yields $a_j = a'_\ell$, after which $\theta_j = \theta'_\ell$. The generic choice of v makes the matching one-to-one, proving $H = H'$. $\square$

The measure representation absorbs the benign Gamma-splitting ambiguity: $a_1\delta_\theta + a_2\delta_\theta$ and $(a_1 + a_2)\delta_\theta$ are literally the same measure. Theorem 1 also makes explicit that the Poisson layer is lossless for population identification, because the count probability-generating function reveals the Laplace transform of the latent intensity distribution [Teicher, 1960].

For the mixture geometry, let

$$P_G = \sum_{s=1}^S \pi_s \text{Gam}(H_s) \quad (42)$$

be the distribution of the latent intensity and define

$$\mathcal{U}(G) = \text{supp}(P_G), \quad \mathcal{C}(G) = \text{cone}\{\mathcal{U}(G)\}.$$

The set $\mathcal{U}(G)$ retains the union of the closure cones of the latent submanifolds, whereas $\mathcal{C}(G)$ is their global conic hull.

Theorem 2 (Identification of support geometry and extreme rays). *Let G and G' be finite probability measures of the form (37). If*

$$\text{MiSo}(G) \stackrel{d}{=} \text{MiSo}(G'),$$

then

$$P_G = P_{G'}, \quad \mathcal{U}(G) = \mathcal{U}(G'), \quad \mathcal{C}(G) = \mathcal{C}(G').$$

Consequently, the extreme rays of the global closure cone are identifiable despite the Poisson observation noise. If rays are normalized by their intersection with $\Delta^{M-1} = \{x \in \mathbb{R}_+^M : \mathbf{1}^\top x = 1\}$, the corresponding extreme points are identifiable as well. Moreover, every global extreme ray contains an atom from at least one motif measure in the support of G and from at least one motif measure in the support of G' .

Proof. The count probability-generating function is the Laplace transform of P_G evaluated at $\mathbf{1} - z$. Equality of the MiSo laws therefore gives equality of the multivariate Laplace transforms of P_G and $P_{G'}$ on $(0, 1)^M$. Uniqueness of the Laplace transform on $\mathbb{R}_+^M$ gives $P_G = P_{G'}$.

For $H = \sum_j a_j \delta_{\theta_j}$, the independent Gamma coefficients in (36) have joint topological support $\mathbb{R}_+^J$. Their linear image consequently has support

$$\text{supp}\{\text{Gam}(H)\} = \left\{ \sum_j x_j \theta_j : x_j \geq 0 \right\} = \mathcal{C}(H).$$

Because the mixture is finite and every mixture weight is positive,

$$\mathcal{U}(G) = \bigcup_{H \in \text{supp}(G)} \mathcal{C}(H). \quad (43)$$

The same identity holds for G' . Equality of P_G and $P_{G'}$ therefore identifies the union of closure cones and its global conic hull.

Extreme rays and normalized extreme points are intrinsic properties of this identified closed cone. Finally, $\mathcal{C}(G)$ is generated by the finite union of the atoms in the component measures. Every extreme ray of a finitely generated cone contains a vector from any finite generating set; otherwise a point on that ray would be a nontrivial conic combination of vectors off the ray. Applying this fact to the atoms under G and under G' proves the final claim. $\square$

When each factor row is normalized by $\mathbf{1}^\top F_k = 1$, the scaled atom $\theta = F_k/b$ lies on the same ray as F_k , and simplex normalization maps it back to F_k . Theorem 2 therefore identifies the factor rows corresponding to global extreme rays. It does not, by itself, identify interior dictionary rows or assign the identified rays to individual motifs.

Remark 1 (Interior rays prevent unrestricted mixture identification). Theorem 1 makes the single-submanifold kernel injective, but it does not imply that an unrestricted mixture G is identifiable. To see this with all Gamma shapes equal to one, choose linearly independent $\theta_A, \theta_B \in \mathbb{R}_+^M \setminus \{0\}$, fix $\rho \in (0, 1)$, and introduce the interior direction

$$\theta_C = \rho\theta_A + (1 - \rho)\theta_B.$$

Let

$$H_{AB} = \delta_{\theta_A} + \delta_{\theta_B}, \quad H_{BC} = \delta_{\theta_B} + \delta_{\theta_C}, \quad H_{AC} = \delta_{\theta_A} + \delta_{\theta_C}.$$

Writing $A(t) = 1 + \langle \theta_A, t \rangle$ and defining $B(t)$ and $C(t)$ analogously, we have $C = \rho A + (1 - \rho)B$ and hence

$$\begin{aligned} \Phi_{H_{AB}}(t) &= \frac{1}{A(t)B(t)} \\ &= \frac{\rho}{B(t)C(t)} + \frac{1 - \rho}{A(t)C(t)} \\ &= \rho\Phi_{H_{BC}}(t) + (1 - \rho)\Phi_{H_{AC}}(t). \end{aligned} \tag{44}$$

Thus the distinct mixing measures

$$G = \delta_{H_{AB}}, \quad G' = \rho\delta_{H_{BC}} + (1 - \rho)\delta_{H_{AC}}$$

produce exactly the same noisy count law. Geometrically, one latent submanifold has been represented by a mixture whose two closure cones subdivide the original cone along an interior ray. This does not contradict Theorem 2: both representations have the same latent support and the same global extreme rays. Rather, it shows why an over-specified interior factor can prevent identification of the individual motifs even when the global geometry is identifiable.

To state the condition needed for full mixture identification, call a transform family $\{\Phi_H : H \in \mathcal{H}_0\}$ finitely linearly independent if, for every finite collection of distinct $H_1, \dots, H_r \in \mathcal{H}_0$,

$$\sum_{j=1}^r c_j \Phi_{H_j}(t) = 0 \text{ for every } t \in (0, 1)^M \implies c_1 = \dots = c_r = 0. \tag{45}$$

Theorem 3 (Finite-mixture identification). *Let G and G' be finite probability measures on $\mathcal{H}_0$. If $\{\Phi_H : H \in \mathcal{H}_0\}$ is finitely linearly independent, then*

$$\text{MiSo}(G) \stackrel{d}{=} \text{MiSo}(G') \implies G = G'.$$

Conversely, if the transform family is not finitely linearly independent, then the class of finite probability mixtures on $\mathcal{H}_0$ is not identifiable.

Proof. Take the union of the finite supports of G and G' and denote its distinct elements by $H_1, \dots, H_r$. Equality of the noisy count laws and (40) gives

$$\sum_{j=1}^r \{G(H_j) - G'(H_j)\} \Phi_{H_j}(t) = 0, \quad t \in (0, 1)^M.$$

Finite linear independence forces every coefficient to vanish, so $G = G'$. Conversely, consider a nonzero finite linear relation among distinct transforms. At $t = 0$, every $\Phi_H(0) = 1$, so the positive and negative coefficient sums are equal. Separating the two sides and normalizing by their common sum produces two different probability mixing measures with the same count probability-generating function, and hence the same MiSo law. $\square$

Theorem 3 is the standard finite-mixture criterion of Yakowitz and Spragins [1968], within the broader identifiability program of Teicher [1960, 1963]. It separates two questions that are sometimes conflated: injectivity of each component kernel, established by Theorem 1, and linear independence of the whole component family, which is required to identify G .

Corollary 4 (Common simplicial dictionary). *Fix linearly independent vectors $\theta_1, \dots, \theta_K \in \mathbb{R}_+^M \setminus \{0\}$ with $K \leq M$, write $\Theta = \{\theta_1, \dots, \theta_K\}$, and consider*

$$\mathcal{H}_\Theta = \left\{ H_\alpha = \sum_{k=1}^K \alpha_k \delta_{\theta_k} : \alpha \in \mathbb{R}_+^K \setminus \{0\} \right\}, \quad (46)$$

where a zero coordinate means that the corresponding atom is omitted. Then every finite MiSo mixing measure G on $\mathcal{H}_\Theta$ is identifiable.

Proof. For $H_\alpha \in \mathcal{H}_\Theta$,

$$\Phi_{H_\alpha}(t) = \prod_{k=1}^K (1 + \langle \theta_k, t \rangle)^{-\alpha_k}.$$

Because the θ_k are linearly independent, the map $t \mapsto (\langle \theta_1, t \rangle, \dots, \langle \theta_K, t \rangle)$ has full row rank and maps an open subset of $(0, 1)^M$ onto an open subset of $\mathbb{R}^K$. Make the coordinate change

$$y_k = \log(1 + \langle \theta_k, t \rangle).$$

A finite linear relation among transforms associated with distinct $\alpha^{(j)}$ becomes

$$\sum_j c_j \exp\{-\langle \alpha^{(j)}, y \rangle\} = 0$$

on an open set. Choose a direction v for which the finitely many values $\langle \alpha^{(j)}, v \rangle$ are distinct and restrict the identity to a short real line segment parallel to v . Distinct univariate exponentials are linearly independent, so every $c_j = 0$. The result follows from Theorem 3. $\square$

Corollary 4 allows a motif to omit a global ray by setting its corresponding shape to zero. It requires the scaled direction θ_k to be shared across motifs, as occurs when selecting factor row F_k also selects a factor-specific rate β_k . The result also permits a fixed over-specified dictionary with $K' > K$ provided all K' scaled directions remain linearly independent, so necessarily $K' \leq M$. The true model is then embedded by assigning zero shape to the extra directions. However, the corollary conditions on a fixed dictionary; it does not by itself establish joint identifiability when the dictionary is learned.

5 Experiments Needed

The empirical goal is not only to show good reconstruction error. The paper needs to show that MiSo recovers meaningful factor-usage motifs, clusters observations by motif, learns factors when F is unknown, and quantifies uncertainty in settings where hard sparse supports are unstable.

5.1 Baselines

The main baselines should be:

1. Poisson NMF followed by clustering of normalized loading scores;
2. row-wise Poisson SuSiE with the frozen NMF dictionary, followed by clustering of posterior expected loading scores;
3. sparse NMF baselines with tuned sparsity penalties, if available;
4. ablations of MiSo with fixed versus learned F .

The first two baselines are already implemented in the current benchmark code.

5.2 Metrics

The simulation tables should include:

1. factor recovery, measured by cosine matching between fitted rows of F and true rows of F_0 ;
2. motif clustering accuracy, measured by matching $\arg \max_s \omega_{is}$ to true groups;
3. support recovery, measured by converting posterior motif scores to predicted active factors;
4. soft clustering diagnostics, based on group-level averages of ω_{is} rather than only hard labels;
5. posterior uncertainty summaries for $\bar{\gamma}_{sd}$, including entropy, top-two gaps, and targeted pair-wise split metrics;
6. selected motif dimensions $\hat{D}_s$ from the ARD prior means;
7. predictive fit, such as held-out Poisson log likelihood or deviance.

5.3 Core simulation settings

The current roadmap suggests the following simulation studies.

Correctly specified block motifs. Use correct K and nominal D , with groups occupying simple single-factor and multi-factor supports. This verifies that the method can recover factors, clusters, and supports in a clean setting. Current evidence shows feasibility, but a larger seed grid is needed before claiming robust superiority over baselines.

Tree/no-anchor motifs. Use motifs such as $(1, 2, 3)$, $(1, 2, 4)$, $(1, 5, 6)$, $(1, 5, 7)$, $(1, 2)$, and $(1, 5)$, so that no group is explained by a single anchor factor. This is the strongest current positive setting: in a small multi-seed pilot, MiSo has substantially better support recovery than the baseline pipelines.

Overspecified K . Generate data with K true factors but fit with extra candidate factor rows, including near-duplicates of true factors. The key diagnostic is whether $\bar{\gamma}_{sd}$ splits posterior mass across duplicate rows when they are statistically interchangeable. Global entropy is not sufficient; the paper should report pair-specific posterior split metrics for the duplicated factors.

Correlated factors. Generate correlated factor rows at controlled strengths. This tests whether MiSo can express targeted ambiguity between correlated factors. The current evidence suggests that raw entropy can be misleading because weak dimensions may become globally diffuse. The final paper version should sweep correlation strength and report targeted pair uncertainty, not only mean entropy.

Overspecified D . Fit each motif with a nominal dimension larger than the true motif dimension. The target behavior is that extra dimensions either have small ARD prior mean $\alpha_{sd}^0/\beta_{sd}^0$ or small posterior loading mass. The current initialization uses the top D center factors as distinct alignment anchors and averages the aligned row-wise Poisson SuSiE posteriors. A full paper needs multi-seed threshold sweeps for $\hat{D}_s$ and comparisons against the true motif dimensions after matching fitted motifs to groups.

5.4 Real data

The real-data analysis should demonstrate three advantages of the MiSo representation beyond reconstruction quality. First, it should identify recurring motif types: scientifically meaningful combinations of factors that are reused across observations. A small atlas of these motifs, together with their responsibilities, should provide a more concise summary than a typical admixture bar plot with many separately colored factors. Second, it should show that a motif represents a low-dimensional submanifold rather than merely a cluster. In the single-cell application, cells assigned to the same motif should populate its posterior dimension coordinates continuously, with related cell states, marker expression, and—where available—spatial position varying smoothly along a trajectory consistent with differentiation. Third, the analysis should recover structure among the factors themselves. Factor co-use across motifs induces a graph in which broadly reused factors may act as hubs and more specialized factors as leaves; these relationships should be checked against external biological interpretation.

These are goals for the final analysis rather than claims established by the current experiment. The identifiability theory remains to be completed, and the post-processing needed when K , S , and D are over-specified still requires further study. In particular, the current Wasserstein dendrogram and merging rule are exploratory and should not yet be presented as the final procedure.

A concise empirical demonstration. The main real-data result can be organized as one multi-panel figure. The first panel should show the conventional NMF loading bar plot and its loss of readability when K is moderately large. The second should replace it with a small motif atlas, displaying the factors and effective dimension used by each recurring motif together with its prevalence. The third should highlight one well-supported two- or three-dimensional motif and plot its cells in the corresponding posterior dimension coordinates. Cell annotations should be used only after fitting: related states should occupy a connected trajectory, and independently chosen marker genes should vary smoothly along the resulting MiSo coordinate. The fourth panel should show the factor co-use graph, with a small number of biologically interpretable hub, branch, and leaf factors.

A companion table should report only the quantities needed to support this picture: held-out deviance relative to Poisson NMF, the average number of active factors per cell, the effective dimensions of the highlighted motifs, stability across random seeds, and external agreement with cell annotations or an independently estimated pseudotime. Thus the intended claim is not that MiSo improves reconstruction at all costs, but that it preserves predictive fit while replacing a high-dimensional loading display by recurring motifs, within-motif coordinates, and a factor-relation

graph. The primary display should use fitted motifs before any uncertain merging step; alternative post-processing rules can be reported separately as a sensitivity analysis.

5.5 Ablations and robustness checks

The paper should include ablations for:

1. fixed versus learned F ;
2. sensitivity to the number of motifs S ;
3. sensitivity to nominal D and the ARD threshold τ ;
4. sensitivity to duplicate-factor and correlated-factor strength;
5. stability across random seeds.

These experiments are needed to distinguish the proposed statistical model from an optimization artifact.

6 Discussion

MiSo reframes sparse NMF as a motif discovery problem. Instead of estimating sparse loadings independently for each observation, it pools repeated sparsity patterns through latent motifs. The posterior motif probabilities $\bar{\gamma}_{sdk}$ provide an interpretable uncertainty summary over the factor identities defining each motif, while the responsibilities ω_{is} cluster observations by motif. The empirical-Bayes prior means $\alpha_{sd}^0/\beta_{sd}^0$ provide a natural ARD-style diagnostic for the effective dimension of each motif.

The fitting pipeline separates the roles of its components: Poisson NMF estimates the initial dictionary, fixed-dictionary row-wise Poisson SuSiE quantifies observation-level factor uncertainty, and MiSo pools that information into recurring motifs. This modular separation avoids interleaving dictionary estimation with observation-level support learning.

The current evidence supports the promise of this approach, especially in tree/no-anchor simulations and in overfitted-factor uncertainty diagnostics. The main open issues are stable dimension selection, broader multi-seed benchmarks, model-selection guidance for K , S , and D , and a real-data example with scientifically interpretable motifs.

A Threshold-and-recycle initialization

For reproducibility, we record an earlier initialization rule that is retained in the software for historical ablations. Using the motif center c_s and factor ordering r_s defined in the main text, it first forms the ordered set

$$A_s(\tau_{\text{init}}) = (a_{s1}, \dots, a_{sq_s}) = \{r_s(j) : c_{s,r_s(j)} \geq \tau_{\text{init}}\}.$$

If this set is empty, it is replaced by $A_s = (r_s(1))$. The rule then fills the D dimensions cyclically,

$$\kappa_{sd}^{\text{recycle}} = a_{s, 1 + ((d-1) \bmod q_s)}, \quad d = 1, \dots, D.$$

When $q_s = 1$, every dimension is assigned the same factor. Conditional on motif s , the resulting factor-space loading is

$$L_i = \sum_{d=1}^D \lambda_{id} e_{a_{s1}} = \left(\sum_{d=1}^D \lambda_{id} \right) e_{a_{s1}},$$

which lies on a one-dimensional ray regardless of the nominal value of D . More generally, recycling constrains the initialization to at most q_s distinct directions. This can create an artificially low-dimensional starting point when only a few center weights exceed τ_{init} .

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